ADOPTION OF VIRTUAL TRY-ON TECHNOLOGY FOR ONLINE APPAREL SHOPPING

JIYEON KIM AND SANDRA FORSYTHE

Virtual try-on technology (referred to in this article as Virtual Try-on) can deliver product information that is similar to the information obtained from direct product examination. In addition, the interactivity and customer involvement created by Virtual Try-on can enhance the entertainment value of the online shopping experience. We used focus group interviews and an online national survey to investigate online apparel shoppers' use of Virtual Try-on to reduce product risks and increase enjoyment in online shopping. We also examined the impact of two important external variables (innovativeness and technology anxiety) that are not included in the electronic Technology Acceptance Model (e-TAM) but were expected to influence adoption of Virtual Try-on and whether or not gender differences existed in the Virtual Try-on adoption process. We examined this dual (functional and hedonic) role of Virtual Try-on by applying a modified e-TAM model to the Virtual Try-on technology adoption process and tested model invariance among male and female shoppers using Virtual Try-on in a simulated online shopping experience. The extended research model was validated in the context of Virtual Try-on adoption. Technology anxiety and innovativeness had significant moderating effects on the relationship between attitude and use of Virtual Try-on technology; however, there was no significant gender difference in the overall adoption process for Virtual Try-on.

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INTRODUCTION

Decisions to shop in traditional and online shopping channels are influenced by both functional motives (making a good choice) and hedonic motives (enjoyment of the shopping process) (Childers et al., 2001; Babin, Darden, & Griffen, 1994). Many retailers are turning to advanced product visualization technologies that provide sensory input in the online shopping environment to reduce product risks, enhance the entertainment value of the online shopping process, or both. One technology that has been gaining popularity in online apparel shopping sites is Virtual Try-on. With this technology, shoppers can create their own virtual models based on their measurements, facial characteristics, hair color, and body shape. Virtual Try-on allows shoppers to zoom in on product features, rotate and view the product from different angles, and view the product in a variety of colors on a virtual model created to imitate their appearance. Virtual Try-on can deliver product information that is similar to the information obtained from examining the product directly. In addition, the interactivity and customer involvement created by using this technology can enhance the entertainment value of the online shopping experience. Virtual Try-on is becoming more widely available in the online retail environment because this dynamic product imagery decreases the gap in online and offline shopping experiences (Wagner, 2007). Adding interactive technologies such as Virtual Try-on to online shopping sites significantly increases conversions (Wagner, 2007), demonstrating the importance of consumer acceptance of these technologies in maximizing online revenue.

Although Virtual Try-on technology has the potential to affect online shopping significantly, there is virtually no published research examining consumer adoption of this technology. We use the electronic Technology Acceptance Model (e-TAM) to investigate online shoppers' adoption of Virtual Try-on for online shopping. Furthermore, we extend the e-TAM by adding two important variables—technology anxiety and innovativeness—treating them as external variables, directly affecting the intended use of Virtual Try-on, and as moderating variables, affecting the relationship between the attitude and use of Virtual Try-on. Because previous research has shown gender differences in trial and use of technology (Meyers-Levy & Maheswaran, 1991; Venkatesh & Morris, 2000), there may be gender differences in the adoption process of Virtual Try-on. Therefore, it is important to investigate whether or not gender differences exist in the Virtual Try-on acceptance process. We first examine the dual (functional and hedonic) roles of Virtual Try-on technology for the empirical validation of the proposed model. Then, gender differences in the Virtual Try-on adoption process are examined by testing the model invariance for male and female shopper groups as well as path-by-path hypotheses to detect any gender differences in the adoption process.

Two unique contributions of this study include an examination of (1) the direct and moderating roles of technology anxiety and innovativeness of shoppers on Virtual Try-on acceptance and (2) the equivalence of the hypothesized model across genders. The approach used here, using the multiple-group structural equation modeling, allows us to test the variance in Virtual Try-on adoption, adding to the explanatory and predictive power of the e-TAM. Most studies based on TAM test only a master model, which may increase the error of generalization when the results are applied to the public. Invariance testing of the model fit across gender groups can provide important insights in understanding the adoption process of Virtual Try-on by men and women. The examination of both the direct impact of technology anxiety and innovativeness on the use of Virtual Try-on and their moderating roles in attenuating or strengthening the effects of attitude on Virtual Try-on usage may increase the explanatory and predictive power of the proposed model further.

THEORETICAL BACKGROUND

Diffusion of innovation theory explains the process by which an innovation is introduced, communicated, evaluated and, consequently, either adopted or rejected. Most individuals first try a new technology on a partial basis, and only if they perceive advantages in using it will they adopt the innovation (Rogers, 1995). Within the adoption literature, perceptions of innovation characteristics such as relative advantage (usefulness and entertainment value) and complexity (ease of use) (Rogers, 1995; Venkatraman & Price, 1990) and also differences in individual shoppers (e.g., gender and levels of innovativeness and technology anxiety) Meuter et al., 2005; Meuter et al., 2003; Robinson, Marshall, &

Stamps, 2004; Peck & Childers, 2003) have been shown to predict adoption behaviors. Hoffman & Novak (1996) found that anxiety regarding the ability to perform a behavior was related negatively to actual usage behavior. Additional research has shown that respondents with higher levels of technology anxiety use fewer self-service technologies, and that innovativeness influences the way new technologies are perceived (Meuter et al., 2003).

The TAM enjoys wide support as a tool for investigating and predicting user acceptance of information technology (e.g., Chau, 1996; Davis, 1989; Pavlou, 2003; Taylor & Todd, 1995). Based on the theory of reasoned action (TRA), TAM focuses on the role of ease of use and usefulness in predicting attitudes toward using a new technology (Davis, 1989). The enjoyment construct was added to TAM to explain the role of intrinsic motivation in the adoption of a new technology (Davis, Bagozzi, & Warshaw, 1992; Heijden, 2004). Heijden (2000) adapted the original TAM for a Web site context, proposing the eTAM framework, and found the concepts of perceived relative usefulness and perceived relative enjoyment were strong influential variables to usage (Figure 1). Whereas perceived usefulness and perceived enjoyment are strong indicators of Web site revisit intentions, perceived ease of use indirectly affects Web site revisit intentions by influencing the perceived relative usefulness and perceived relative enjoyment (Heijden, 2000). The eTAM model is consistent with previous research on retail shopping behavior and supports the presence of both utilitarian and hedonic motivations for online shopping (Babin et al., 1994; Childers et al., 2001).

Within the eTAM framework, perceived usefulness of a technology reflects functionality, and enjoyment reflects hedonic aspects of the online shopping process. While some consumers may use Virtual Try-on technology primarily for functional purposes, such as improved multidimensional examination of a product, others may use it primarily for hedonic purposes (cf. Childers et al., 2001), by creating a virtual model and viewing various apparel items on the model. As online shoppers find Virtual Try-on technology to be effective in reducing product risk, increasing shopping enjoyment, or both, they will be more likely to adopt it.

RESEARCH MODEL AND HYPOTHESES

The proposed model of Virtual Try-on acceptance extends the e-TAM and explains the adoption process of Virtual Try-on for online apparel shopping. The proposed model allows us to examine (1) the relationships between perceived usefulness, ease of use, and entertainment value of Virtual Try-on; (2) the influence of these beliefs (perceived usefulness, ease of use, and entertainment value) on attitudes toward using Virtual Try-on; (3) the relationship between the attitudes toward using Virtual Try-on and intended use of Virtual Try-on; (4) post-use evaluation of Virtual Try-on for online

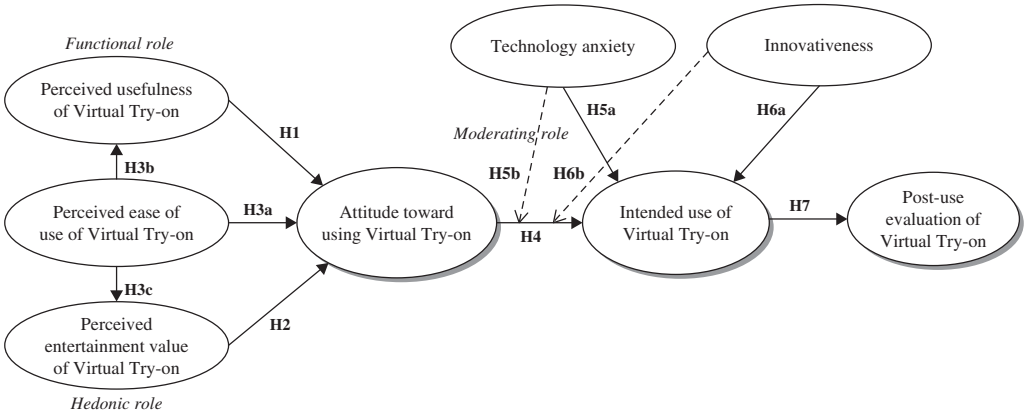


FIGURE 1

Virtual Try-on Acceptance Model

apparel shopping; and (5) direct and moderating effects of technology anxiety and innovativeness.

In the proposed model, perceived ease of use influences shoppers' attitudes toward using Virtual Try-on and indirectly influences their attitudes through its influence on perceived usefulness and entertainment. Shoppers' positive attitudes toward using Virtual Try-on are expected to influence their intended use of Virtual Try-on favorably. Most TAM-based research examining users' technology acceptance has not gone beyond the behavior or behavioral intention of using the system/technology, but to examine the true acceptance of Virtual Try-on, it is important to consider post-use evaluation of the apparel shopping experience aided by Virtual Try-on technology as well. One exception is Kim & Forsythe's (2007) examination of functional and hedonic roles of Virtual Try-on technology. Dabholkar & Bagozzi (2002) examined the moderating effects of some consumer traits, such as novelty seeking, self-efficacy, and need for interaction on the relationship between beliefs (ease of use, performance, and fun) and attitude towards using technology-based self-service. It seems likely that additional consumer traits, such as technology anxiety and innovativeness, may affect intended use of Virtual Try-on directly regardless of the shopper's attitude toward using Virtual Try-on. However, the role of technology anxiety and innovativeness on Virtual Try-on usage has not been examined yet.

Based on the proposed adoption model, research hypotheses are posited regarding the relationships between perceived usefulness, perceived ease of use, perceived entertainment value, attitude, use, and post-use evaluation, as well as the direct and moderating effect of innovativeness and technology anxiety on the use of Virtual Try-on. In addition, gender differences in the Virtual Try-on adoption process are examined.

H1. Perceived usefulness of Virtual Try-on will have a positive influence on attitudes towards using Virtual Try-on.

Insufficient information on product attributes and shoppers' inability to accurately evaluate the quality of the product in the online environment results in increased product risk. Online shoppers can use Virtual Try-on to better evaluate the online product prior to purchase, thereby reducing the likelihood of a poor choice. Online apparel shoppers are concerned about making poor decisions due to inability to examine the product physically, especially given that many of the characteristics of apparel that are important in consumer decision making, such as appearance on the body and fit, are difficult to present onscreen and standard descriptors of a product often are insufficient for product evaluation (Grewal, Iyer, & Levy, 2004; Kartsounis, Magnenat-Thalmann, & Rodrian, 2001). Therefore, using Virtual Try-on as a proxy for physical examination may be especially important when shopping for apparel (Citrin et al., 2003) because it provides proxy sensory experiences that can serve as a surrogate for direct product examination when evaluating apparel products online.

H2. Perceived entertainment value of Virtual Try-on will have a positive influence on attitudes towards using Virtual Try-on.

The entertainment provided by shopping has been found to be an important motivator both in traditional shopping environments (Bloch, Sherrel, & Ridgway, 1986; Babin et al., 1994) and online (Hoffman & Novak, 1996; Childers et al., 2001). Trying clothing on a virtually created model can provide entertainment in addition to facilitating product evaluation. Given that hedonic use of the Internet plays an important role in online shopping (Childers et al., 2001; Menon & Kahn, 2002), the entertainment value provided by interaction with Virtual Try-on technology is likely to create more positive attitudes toward using Virtual Try-on when shopping for apparel online.

H3a. Perceived ease of use of Virtual Try-on will have a positive influence on attitudes toward using Virtual Try-on.

Research has confirmed that ease of use is an important factor in predicting attitudes toward technology-based self-service (Dabholkar, 1994; Davis et al., 1992, Heijden, 2000). According to Rogers (1995), complexity, the antithesis of ease of use (Agarwal & Prasad, 1997), reduces an individual's willingness to adopt a system. Previous researchers found that perceived ease of use had a positive influence on users' attitudes towards using the Internet to send e-mail (Szajna, 1996; Gefen & Straub, 1997). Liao et al. (1999) reported that the easier it is to use an Internet banking service, the more positive the attitude toward using this service. Therefore, perceived ease of use is expected to have a positive effect on consumer attitudes towards using Virtual Try-on.

H3b. Perceived ease of use of Virtual Try-on will have a positive influence on perceived usefulness of Virtual Try-on.

Previous research demonstrates strong empirical support for a positive relationship between perceived ease of use and perceived usefulness (Davis, 1989; Adams, Nelson, & Todd, 1992; Segars & Grover, 1993). Thus, the easier Virtual Try-on is to use, the more useful it will be perceived to be (Heijden, 2000).

H3c. Perceived ease of use of Virtual Try-on will have a positive influence on the perceived entertainment value of Virtual Try-on.

Igbaria, Parasuraman, & Baroudi (1996) found support for a positive relationship between perceived entertainment value and system usage. By contrast, perceived complexity (the opposite of ease of use) was correlated negatively with perceived entertainment value (Igbaria, Parasuraman, & Baroudi, 1996). These findings lead to the expectation that the easier Virtual Try-on is to use, the greater the perceived entertainment value for online shopping.

H4. Attitudes towards using Virtual Try-on will have a positive influence on intended use of Virtual Try-on.

The innovation literature specifies that an individual's attitude toward using an innovation influences adoption of the innovation (Rogers, 1995). Therefore, an individual's use of a technology is a function of his or her attitude toward its use (Moore & Benbasat, 1991). The theory of reasoned action, on which TAM is based, suggests that the more positive the attitude toward performing a behavior, the more likely an individual is to perform that behavior (Ajzen & Fishbein, 1980). Consumers who have favorable attitudes toward online shopping are less likely to abort intended transactions (Cho, 2004). Thus, consumers who have a positive attitude toward using Virtual Try-on are expected to be more likely to use Virtual Try-on for online apparel shopping.

H5. Technology anxiety will (1) have a negative impact on intended use of Virtual Try-on technology and (2) moderate the impact of attitude toward using Virtual Try-on on actual usage.

Online shoppers are not likely to use Virtual Try-on (even when they see its benefits) unless they feel comfortable with the technology. Ajzen (1991) asserted that behavior is strongly influenced by perceived ability to perform that behavior, while Rogers (1995) noted that people are more likely to adopt an innovation with which they are comfortable. Technology anxiety, the fear and apprehension people feel when thinking about or actually using technology-related tools (Cambre & Cook, 1985; Meuter et al., 2003; Scott & Rockwell, 1997) is expected to influence the use of Virtual Try-on directly as well as to moderate the relationship between the attitude and the use of Virtual Try-on.

H6. Consumers' innovativeness will (a) have a negative impact on intended use of Virtual Try-on technology and (b) moderate the impact of attitude toward using Virtual Try-on on actual usage.

Innovativeness, the latent underlying preference for new and different experiences (Hirschman, 1980), motivates a search for new experiences that stimulate the mind, the senses, or both (Pearson, 1970; Hirschman, 1984; Venkatraman & Price, 1990). In a technology context, *innovativeness* is defined as the willingness of an individual to try out new technology (Robinson, Marshall, & Stamps, 2004). For example, adoption of in-home shopping methods is not only a function of attitudes, needs, and experiences, but also personal characteristics such as innovativeness (Eastlick, 1993; Robinson, Marshall, & Stamps, 2004; Shim & Drake, 1990). Consumers with high levels of innovativeness are more likely to seek multiple sources of information and new experiences that stimulate their senses as they form their perceptions about a technology (Agarwal & Prasad, 1998; Flynn & Goldsmith, 1993; Midgley & Dowling, 1978; Hirschman, 1984). Therefore, innovativeness is expected to influence the use of Virtual Try-on positively and moderate the relationship between attitude toward using Virtual Try-on and the intended use of Virtual Try-on.

H7. Use of Virtual Try-on will result in a positive evaluation of Virtual Try-on for online apparel shopping.

Post-use evaluation is defined as an individual's subjectively derived evaluation of any outcome and/or experience associated with using technology (Westbrook, 1980). Individuals will be more likely to adopt a specific behavior if they perceive that it will lead to positive outcomes (Compeau & Higgins, 1995). If their experience with using Virtual Try-on technology is

satisfactory, shoppers are likely to evaluate Virtual Try-on favorably. Schlosser (2003) found that both browsers and searchers reported more positive attitudes and purchase intentions after visiting a Web site with interactive virtual product presentations. Therefore, intended use of Virtual Try-on is expected to result in positive post-use evaluations.

H8. The adoption process of Virtual Try-on for online apparel shopping will differ by gender.

Previous research suggests that men and women differ with respect to information processing and have different levels of technology self-efficacy (Venkatesh & Morris, 2000). Men and women also differ with regard to characteristics that they consider important in evaluating products and in information-processing strategies (Meyers-Levy & Maheswaran, 1991). Thus, gender difference may exist in perceptions of the functional and hedonic roles of Virtual Try-on, attitude toward using Virtual Try-on technology, and innovativeness and technology anxiety. Therefore, the adoption process for Virtual Try-on technology in online apparel shopping may differ by gender.

METHOD

Focus Group Interviews

Five focus group interviews were conducted in lab settings to understand how the experience of different product visualization technologies can (1) help purchase decision making by providing product information, (2) provide entertaining shopping experiences, (3) identify the types of product visualization technologies consumers find most effective and useful for particular products, and (4) examine any gender difference in perceptions of using the technology. Students majoring in retail management and fashion merchandising at a large university in the southeastern United States were recruited for the interview. Ten students participated in each focus group interview; the participants were 20% male and 80% female. A brief oral presentation about the Virtual Try-on technology was given to the participants prior to their participation in a simulated online shopping activity using the technology. After the interviewees completed the simulation, the researcher asked them to talk about the usefulness, ease of use, and entertainment value of the technologies to stimulate discussion based on their simulated shopping experience. The researcher also asked about the type of apparel products for which Virtual Try-on technology would be most helpful, to identify practical applications for online apparel retailers.

Online Survey

An online survey was developed to examine the acceptance of Virtual Try-on by male and female online apparel shoppers. The constructs were measured using 7-point Likert-type scales ranging from 1 (strongly disagree) to 7 (strongly agree). A total of 36 items were used to measure the eight latent constructs: technology anxiety (TA), innovativeness (INN), perceived usefulness (PU), perceived ease of use (PEOU), perceived entertainment value (PE), attitude (ATT), intended use (USE), and post-use evaluation (EVA). Conceptual definitions and sources are presented in Table 1.

An online survey was administered to a national panel of online shoppers randomly selected from a pool of participants included in a database purchased from a commercial survey provider. Selected members of the panel received an e-mail containing the online survey link. Upon clicking the link provided in the survey, they were led to the stimulus retail Web site with Virtual Try-on technology. The participants were asked to create their own virtual model and try on a top, an outerwear item, and a pair of pants. After completing the online apparel shopping simulation using Virtual try-on, respondents completed the survey questions regarding their simulated shopping experience using Virtual Try-on technology.

There were 491 valid and complete responses from the 2,000 online survey invitees, for a 25% response rate. Of these, 54% were male and 46% were female. A total of 13% of the respondents had not purchased apparel online during the past six months, 41% had done so 1–2 times, 23% 3–4 times, 15% 5–6 times, and the remaining 8% had purchased apparel online more than 6 times. And 35% of the respondents had spent a total of $1 to $100, 25% spent $101 to $200, 13% spent $201 to $300, 6% of spent $301 to $400, and the rest (9%) spent $401 or more on apparel purchases online during the past six months. The sample included shoppers with very little to extensive

TABLE 1 Conceptual Definitions and Sources of Constructs

CONSTRUCT	CONCEPTUAL DEFINITION	SOURCE
Technology anxiety	The fear and apprehension people feel when considering use of or actually using (Virtual Try-on) technology-related tools.	Cambre & Cook, 1985; Scott & Rockwell, 1997; Meuter et al., 2003
Innovativeness	In a technology context, innovativeness is the willingness of an individual to try new (Virtual Try-on) technology.	Robinson, Marshall, & Stamps, 2004
Perceived usefulness	The degree to which a person believes that using a (Virtual Try-on) technology would enhance his or her task-related performance.	Davis, 1989
Perceived ease of use	The degree to which a person believes that using a (Virtual Try-on) technology would be free of effort.	Davis, 1989
Perceived entertainment value (perceived enjoyment)	The extent to which the activity of using a (Virtual Try-on) technology is perceived to be enjoyable in its own right, apart from any performance consequences that may be anticipated.	Davis, Bagozzi, & Warshaw, 1992
Attitude	An overall affective evaluation (of using Virtual Try-on) that can range from extremely positive to extremely negative.	Childers et al., 2001
Intended use	Consumers' intended use of (Virtual Try-on) technology when shopping online.	
Post-use evaluation	An individual's subjectively derived evaluation of any outcome and/or experience associated with using (Virtual Try-on) technology.	Westbrook, 1980

online shopping experience and was generally representative of online shoppers. Although 62% of the respondents had used Virtual Try-on before, the chi-square difference test between inexperienced vs. experienced users of Virtual Try-on showed no significant differences ($\Delta\chi^2 = 9.22$, $\Delta df = 7$, $p = .24$) in the overall parameter estimates, indicating the level of experience with this technology was not a main affecting factor.

RESULTS AND DISCUSSION

Focus Group Interview

Interviewees had mixed opinions about using Virtual Try-on for online apparel shopping. Two interviewees found using Virtual Try-on to be confusing, but the rest didn't find it difficult to use. One interviewee said, "I do not know my measurements well, such as my thigh size. I was confused by that matter but not by the procedure of creating the model." Another interviewee said, "It was interesting to create my model, but the clothing didn't look like it would on the real me." Another interviewee said, "It was interesting to create my model and try clothing on it . . . . . more amusing than anything else. It was also useful for putting items together to make a whole outfit." As for their intention to use Virtual Try-on, one interviewee said, "For fun, I will do it, but it doesn't provide me with a whole lot of information . . . . the clothing doesn't look realistic to me. I would recommend it to others if they had some extra time to set the model up to look like them and wanted to play around with outfits, colors, and combinations of clothes." Another said "Virtual Try-on would be useful for online apparel shopping because it shows how the clothing would look on the body. . . . seeing how colors look with my skin and hair colors and seeing if the clothes fit properly on my upper and lower body lengths help." Overall, both male and female interviewees found Virtual Try-on to

be fun to use. The majority of the interviewees agreed that Virtual Try-on technology was more entertaining than functional because the way the clothing looked on the model didn't help that much in showing how it would actually look on them in person. Many interviewees indicated that they would visit online shopping sites providing Virtual Try-on and recommend it to others for trial. Male interviewees noted that men's clothing usually does not have body-close construction. One male interviewee said, "It wouldn't matter much to see the cut and draping of the clothing on my virtual model. I would be fine with the views from different angles without the model." Male interviewees indicated that Virtual Try-on would be useful for buying suits and jeans, whereas female interviewees believed that Virtual Try-on would be helpful when shopping for underwear, bathing suits, or dresses, as it allowed them to see how the item would look like on a body.

Reliability and Validity of Measures

The results of the reliability tests showed that all the construct measures were reliable, with Cronbach alphas greater than 0.8. Principal component analysis showed satisfactory discriminant validity among constructs and good internal consistency for most constructs except for the attitude and use. To ensure that the measures were not measuring the same constructs, discriminant validity tests were performed for these two constructs. First, we constrained the estimated correlation parameter between two measurement models to 1.0, and a chi-square (χ^2) value was generated for the constrained model, imposing a perfect correlation between the two constructs. Then, the chi-square (χ^2) value was generated for the unconstrained model with the free correlation parameter estimation. The chi-square difference test between two models verified the discriminant validity of the two variables-attitude and use. Confirmatory Factor Analyses (CFA) were conducted for all measurement models. CFA item factor loadings for the latent constructs (greater than 0.6), indicated the scale items were a good manifestation of the constructs (Marsh & Hau, 1999). Based on a rule of thumb for the incremental goodness-of-fit indexes (Hu & Bentler, 1999), the model fit for all eight measurement models was good, with all comparative fit index (CFI) and goodness of fit index (GFI) values greater than 0.9.

Structural Model Evaluation and Multiple Group Comparison

The hypothesized model was assessed by maximum likelihood estimation and evaluated by three fit measures—the CFI, the GFI, and the root mean square error of approximation (RMSEA)—using Amos 5.0 (Arbuckle, 2003). According the threshold suggested by researchers (Browne & Cudeck, 1993; Hu & Bentler, 1999), the fit indexes indicated an acceptable model fit for the proposed model across the groups, with CFI = 0.8, GFI = 0.9, and RMSEA = 0.05.

After the initial model assessment for the proposed model, we conducted Structural Equation Modeling (SEM) for male and female groups to test proposed hypotheses by estimating the structural coefficients for the male group and the female group separately. These results are displayed in Figures 2 and 3, respectively. The standardized structural coefficients and

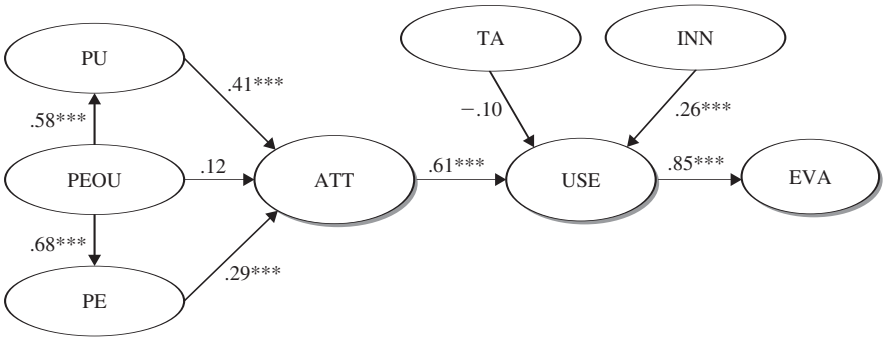


FIGURE 2

Acceptance Model of Virtual Try-on by Female Shoppers

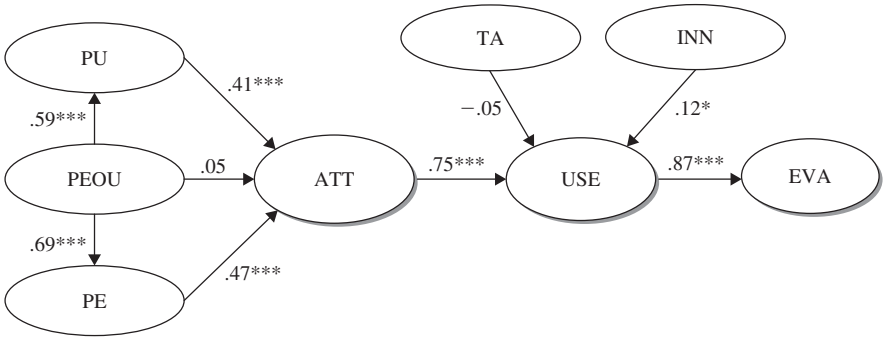


FIGURE 3

Acceptance Model of Virtual Try-on by Male Shoppers

*** = sig. at $p < .001$, ** = sig. at $p < .01$, * = sig. at $p < .05$

the significance are compared within a group and between the two groups in Table 2.

Hypotheses regarding the effects of perceived usefulness and entertainment value on the attitude were supported strongly. These results confirmed perceived usefulness (H1) and perceived entertainment value (H2) as strong predictors of attitudes towards using Virtual Try-on for men and women and are consistent with research supporting the presence of both functional and hedonic motivations in retail shopping (Babin, Darden, & Griffin, 1994) and in online shopping (Childers et al., 2001).

Perceived ease of use showed a positive influence on both perceived usefulness (H3b) and perceived entertainment value (H3c). Interestingly, the direct influence of perceived ease of use on the attitude (H3a) was rejected for both men ($\beta = 0.12$, $p = 0.12$) and women ($\beta = 0.05$, $p = 0.53$). Unlike the other two beliefs (PU and PE), the direct influence of perceived ease of use on the attitude has been inconsistent in previous research, and most authors agree that the influence of perceived ease of use on attitude is mediated by perceived usefulness and enjoyment (Davis, Bagozzi, & Warshaw, 1992). Thus, it seems likely that

TABLE 2 Within-Group Structural Coefficients and Significance for Hypotheses

		MALE		FEMALE	
HYPOTHESES		**COEFFICIENT**	**SIG. P**	**COEFFICIENT**	**SIG.**
H1.	PU → ATT	0.41	***	0.41	***
H2.	PE → ATT	0.29	***	0.47	***
H3a.	PEOU → ATT	0.12	NS ($p = 0.12$)	0.05	NS ($p = 0.53$)
H3b	PEOU → PU	0.58	***	0.59	***
H3c	PEOU → PE	0.68	***	0.69	***
H4	ATT → USE	0.61	***	0.75	***
H5a	TA → USE	−0.10	NS ($p = 0.05$)	−0.05	NS ($p = 0.35$)
H6a	INN → USE	0.26	***	0.12	* ($p = 0.02$)
H7	USE → EVA	0.85	***	0.87	***

*** = sig. at $p < 0.001$, ** = sig. at $p < 0.01$, * = sig. at $p < 0.05$

TABLE 3 Multiple-Group Structural Model Invariance Test

GROUPS	MODEL DESCRIPTION	χ^2	df	$\Delta\chi^2$	Δdf	SIG.	INVARIANT
Male/female	Base model (free estimation)*	2255.94	630	12.15	9	NS ($p = 0.21$)	Yes
	Model with equality constraint imposed	2268.09	639				

*Base model: two-group structural model (male/female)

the influence of perceived ease of use on attitude may differ in different context.

Attitudes towards using Virtual Try-on had a significant influence on the intended use of Virtual Try-on. As expected, intended use of the Virtual Try-on had a positive influence on post-use evaluation. The effect of perceived entertainment value on the attitude (PE $\rightarrow$ ATT) was stronger for women ($\beta = 0.47, p < 0.001$) then for men ($\beta = 0.29, p < 0.001$), supporting the notion that hedonic motivations are more important for women than men when using Virtual Try-on for shopping online (Venkatesh & Morris, 2000). Previous research has indicated that, compared to women, men are more task-oriented (Minton & Schneider, 1980). However, this study showed the effect of perceived usefulness on the attitude (PU $\rightarrow$ ATT) was not different for men ($\beta = 0.41, p < 0.001$) and women ($\beta = 0.41, p < 0.001$).

After the separate research model hypotheses testing for male and female groups, we conducted multiple-group SEM to test H8 regarding gender differences in the overall adoption process of Virtual Try-on. This was achieved by testing the invariance of path parameters across male and female groups simultaneously. The invariance test for the model was achieved by comparing chi-square (χ^2) values and degrees of freedom (df) for the base model and the constrained model. In this comparison, the increase in χ^2 values due to the equality constraints was used as a significance test (Byrne & Campbell, 1999; Byrne, 2001; Kline, 1998; Raju, Lafitte, & Byrne, 2002). All path parameters were constrained to be equal across two groups to test whether the constrained model was invariant between the groups. Then the fit of the base model (free parameter estimation) and the constrained model (equality constraints imposed on parameter estimation) were compared. The summary of the χ^2 values and $\Delta\chi^2$ values (differences of χ^2 values between the base model and constrained model) for the series of analyses involved in testing invariance are presented in Table 3.

The first entry shows the fit (χ^2) of the initially hypothesized structural model when tested simultaneously across two groups with no equality constraints. The second entry reports the fit of the model when equality constraints were imposed on all path parameter estimations. The model fit difference from the comparison of the two groups does not indicate that the coefficients for the male model and the female model were significantly different ($\Delta\chi^2 = 12.15, \Delta df = 9, p = 0.21$). This leads to the conclusion that the adoption process for Virtual Try-on is not significantly different for men and women with respect to overall estimated path parameters. The lack of gender differences in technology acceptance for online shopping differs from some previous research suggesting that women have higher computer anxiety (Hoffman & Novak, 1996; Venkatesh & Morris, 2000). However, since the explosion of Internet shopping in recent years, women are more likely to be familiar with these technologies; thus, their anxiety level is likely to be low regarding this particular technology.

In addition to the direct impact of technology anxiety and innovativeness on intended use of Virtual Try-on, it was also important to examine the significance of these two variables in moderating the positive relationship between the attitude and the intended use of Virtual Try-on. To examine the moderating effects of technology anxiety and innovativeness on the relationship between attitude and use, technology anxiety and

TABLE 4 Moderating Effects of Technology Anxiety and Innovativeness on the Relationship Between Attitude and Use

HYPOTHESES	MODERATING VARIABLES	LEVELS	ESTIMATE (UNSTANDARDIZED COEFFICIENT)	$\Delta\chi^2$	Δdf	SIG. P	MODEL INVARIANCE	MODERATING EFFECT
H5b	Technology anxiety (TA)	High	0.83	20.6	7	0.004	No (variant)	Yes
		Low	1.27					
H6b	Innovativeness (INN)	High	1.28	19.1	7	0.008	No (variant)	Yes
		Low	0.58					

innovativeness factors were divided into high and low groups. Then the structural model with free parameter estimate and the model with an equality constraint imposed on the path between attitude and use (ATT → USE parameter estimate) were tested simultaneously; the worsening in fit measures by the increase in chi-square would verify the significant difference between two models—the high level of technology anxiety group and the low level of technology anxiety group. The same test was conducted for the other two models—the high level of innovativeness group and the low level of innovativeness group. Because the equality constraint was imposed only on the effect of attitude on the intended use of Virtual Try-on, the differences shown by the chi-square difference ($\Delta\chi^2$) verifies the moderating effects of technology anxiety and innovativeness. The changes in the unstandardized coefficients show the levels of changes (high/low TA and high/low INN) in effects of attitude on the intended use of Virtual Try-on, signifying the moderating effects.

The hypotheses regarding the moderating effect of technology anxiety (H5b) and innovativeness (H6b) on the relationship between the attitude and intended use of Virtual Try-on were supported. The moderating effects of both technology anxiety ($\Delta\chi^2 = 20.6$, $\Delta df = 7$) and innovativeness ($\Delta\chi^2 = 19.1$, $\Delta df = 7$) were significant at the significance level of 0.01. This indicates significant overall moderating effects of technology anxiety and innovativeness in the adoption process of Virtual Try-on. The estimated coefficient of the effect of attitude on the use increased from 0.83 (high) to 1.27 (low) for technology anxiety and decreased from 1.28 (high) to 0.58 (low) for innovativeness. In other words, the effect of attitude on use is likely to be more evident (or stronger) for those with low levels of technology anxiety and high levels of innovativeness than those with high level of technology anxiety and low level of innovativeness (see Table 4).

IMPLICATIONS AND LIMITATIONS

The results of this study provide empirical support for perceived usefulness and perceived entertainment value as strong predictors of consumers' attitudes towards using Virtual Try-on for online apparel shopping. There was no significant gender difference in the overall adoption process of Virtual Try-on. Although this result contradicts previous research, these findings suggest the factors affecting the adoption process for online shopping technologies may be changing as women gain experience in using these technologies in the online shopping environment. The technology anxiety and innovativeness had significant moderating effects on the relationship between attitude and use of Virtual Try-on technology. A significant contribution of this study is providing empirical validation of the e-TAM with respect to adoption of Virtual Try-on technology and extending this model to include technology anxiety, innovativeness, and gender for more accurate predictions and improved generalization.

Online clothing sales are growing rapidly and are now greater than online sales for computer hardware. Nevertheless, returns for clothing bought online are twice as high (14%) as for other products bought online, resulting in increased cost of restocking and

reselling the merchandise (Barbaro, 2007). Because shoppers will not easily abandon the habit of trying on clothes, it is important that online clothing retailers more effectively use product visualization technologies (such as Virtual Try-on) to provide better pre-purchase information similar to that obtained in trying on clothing to reduce poor online clothing choices and the expenses associated with returns. The rapid growth of high-speed Internet access (now in 50% of U.S. households) has made it possible for online shoppers to download product images quickly (Barbaro, 2007). Product visualization technologies such as Virtual Try-on may reduce the risk and increase the entertainment value associated with purchasing clothing online, thereby bridging the gap between online and offline clothes-shopping experiences. Thus, gaining a greater depth of knowledge about what drives consumers to use new online technologies such as Virtual Try-on is of considerable interest to retailers.

Our results suggest that although Virtual Try-on contributed to the hedonic dimension of online shopping, most online shoppers lack confidence that Virtual Try-on technology provides reliable information regarding fit and the way a clothing item actually will look on them. Nevertheless, Virtual Try-on technology may aid in promoting multiple sales because shoppers enjoyed putting various items together on the virtual model. Furthermore, the entertainment generated by interacting with the site to build virtual models and experiment with various clothing items may encourage shoppers to spend more time on the site and potentially purchase more items on the site. However, to increase the functionality of Virtual Try-on technology in improving online clothing purchase decisions, it will be important to portray how selected clothing items will look on the online shopper more accurately and to convey this benefit successfully to online apparel shoppers. If Virtual Try-on technology could be combined with other technologies to address shoppers' appearance and fit issues better, the perceived functionality of Virtual Try-on technology would be enhanced greatly.

In fact, with the implementation of three-dimensional (3-D) body scanning technology, product visualization technologies such as Virtual Try-on have potential to revolutionize online apparel shopping. Currently, 3-D scanners, such as the Intellifit scanner, help retailers trying to solve fit problems and facilitate mass customization; furthermore, 3-D body scan technology has been used in cyberspace. For example, MyShape employs a system that matches brands and styles to each user's measurements and style preferences, but it doesn't offer to show the look on a virtual model. Combining these technologies to use virtualization technologies more effectively as a tool to reduce risk and improve online clothing purchase decisions could be the next major step in driving online apparel sales.

Inherent within any study are potential limitations that may affect the overall validity and reliability of the results. One limitation is the use of student samples as the focus group interviewees; students are biased in terms of gender, age, and geographic location. Further generalization could be obtained by including a broader sample. We also acknowledge the potential limitation of the self-completed survey—the online national survey was not conducted in a controlled lab environment. This may restrict external validity and should be kept in mind when interpreting the results. Because a limited number of products were included in the simulation, the findings may not be extrapolated to other product categories. Nonetheless, this study represents an important initial attempt to conceptualize the adoption of this relatively unexplored technology in online apparel shopping.

Replications and extensions of this research would be helpful to understand the impact of Virtual Try-on technology more fully in online retailing. Future studies could examine and compare consumers' perceptions of various product visualization technologies for product evaluation and entertainment in online shopping. The particular demographics associated with the user acceptance, usage patterns, post-purchase satisfaction, and future patronage of a Web site also could be investigated. Acceptance of various online shopping technologies may also be influenced by consumer characteristics such as time-consciousness, opinion leadership (Korgaonkar & Moschis, 1987) or by the shopper's online shopping motivations. Thus, it will be important to identify specific consumer variables that influence the adoption of various product visualization technologies. In particular, we encourage extension of this research to examine further the potential of combining 3-D body scanning technology with product visualization technologies for improved online apparel shopping.

Virtual Try-on technology may have considerable potential in attracting new customers. The female focus group interviewees indicated that they would recommend the site with Virtual Try-on to others, but the male interviewees did not. Thus, it would be worthwhile to see how useful this technology is in attracting new customers and keeping existing customers.

The results of this study are consistent with the findings of other researchers who have reported that enhanced experiences provided by interactive technologies result in stronger purchase intentions than passive product presentations (Li, Daugherty, & Biocca, 2002, 2003; Klein, 2003; Schlosser, 2003) and have important implications for online retailers. Virtual Try-on allows consumers to simulate the functionality and/or appearance of a product, creating a more compelling online shopping experience. By reducing perceived risk and enhancing shopping enjoyment, Virtual Try-on technology has the potential to increase the number of unique and repeat traffic visitors to the site, increase online sales, and ultimately establish an online competitive advantage. Online retailers offering products that particularly benefit from this technology (e.g., underwear, bathing suits, jeans, or dresses) certainly should consider applying a product visualization technology such as Virtual Try-on.

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